Project Objective

Design a complete **Off-grid solar photovoltaic system** to supply a small residential house with reliable electricity, ensuring autonomy from the grid during daytime and nighttime using battery storage.

Load Analysis (Daily Consumption)

Typical Household Loads:

Appliance	Power (W)	Hours/day	Energy (Wh/day)
LED Lights (6 units)	60 W	5 h	300 Wh
Refrigerator	150 W	24 h (40% duty cycle)	1440 Wh
TV	100 W	4 h	400 Wh
Fan (2 units)	120 W	6 h	720 Wh
Laptop + Charging	100 W	4 h	400 Wh

Total Daily Energy Consumption:

= 3260 Wh/day ≈ 3.26 kWh/day

Solar Resource Assumption

- Location: Cairo, Egypt
- Average Peak Sun Hours (PSH): **5.5 hours/day**

PV System Sizing

1. Required PV Power

PV Power=Daily LoadPSH×System EfficiencyPV\ Power = \frac{Daily\ Load}{PSH \times System\ Efficiency}PV Power=PSH×System EfficiencyDaily Load

Assume system efficiency = 0.75

PV=32605.5×0.75=790WPV = \frac{3260}{5.5 \times 0.75} = 790 WPV=5.5×0.753260 =790W

Final PV Size (with safety margin)

1 kW Solar PV System

Panel Selection:

- 2 × 550W Mono-crystalline panels
  Total = **1100 Wp system**

Battery Sizing

Requirement:

- Autonomy: 1 day backup
- System Voltage: 24V
- Battery Type: Lithium or Lead Acid

Battery Capacity=326024×0.85Battery\ Capacity = \frac{3260}{24 \times 0.85}Battery Capacity=24×0.853260

= **160 Ah approx**

With safety margin:

24V – 200Ah Battery Bank

Charge Controller

Current=PV PowerSystem Voltage=110024≈46ACurrent = \frac{PV\ Power}{System\ Voltage} = \frac{1100}{24} ≈ 46ACurrent=System VoltagePV Power =241100 ≈46A

Recommended:

- **MPPT Charge Controller 60A – 24V**

Inverter Selection

Peak Load Estimation:

- Simultaneous load ≈ 1500–1800 W

Recommended inverter:

- **Pure Sine Wave Inverter 2 kW – 24V**

System Configuration

Final System Design:

- 2 × 550W Solar Panels (Series/Parallel depending MPPT range)
- 24V 200Ah Battery Bank
- 60A MPPT Charge Controller
- 2kW Pure Sine Wave Inverter

System Performance

- Daily energy supply: **3.26 kWh**
- Backup autonomy: **1 day**
- System type: **Off-Grid Solar PV System**
- Efficiency: ~75–80%

Engineering Considerations

- Battery depth of discharge (DoD) considered at 80–85%
- System losses included (temperature, wiring, inverter loss)
- Oversizing PV recommended for cloudy days
- MPPT improves efficiency over PWM controllers